DO NOW — RECALL

Paper 1 targeted reteaching

Name: _____ Class: _____ Date: _____

1. What does CIR hold?

2. What is a foreign key?

3. What is DNS used for?

4. Simplify A∨(¬A∧B), showing working.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Current instruction.

2. Model answer

Reference to a key in another table.

3. Model answer

Resolving a domain name to an IP address.

4. Model answer

$(A \vee \neg A) \wedge (A \vee B) = A \vee B$.

By the end, you can…

- Diagnose specific Paper 1 misconceptions
- Correct numerical and technical explanations
- Transfer improved method to new questions

Vocabulary — know these terms

reteach — Address an identified knowledge gap.

transfer — Apply a method to changed values.

method mark — Credit for correct intermediate reasoning.

NEW LEARNING

Read, annotate and explain

Identify

Identify exact lost mark or incorrect idea

Model

Study fresh model with different values

Apply

Try an independent unseen example

Evaluate

Record the correction for exam revision

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

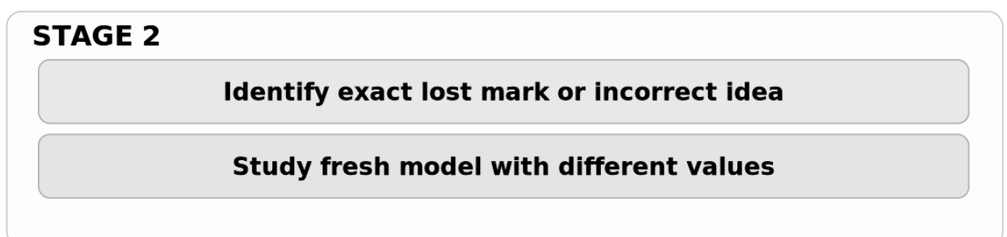
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



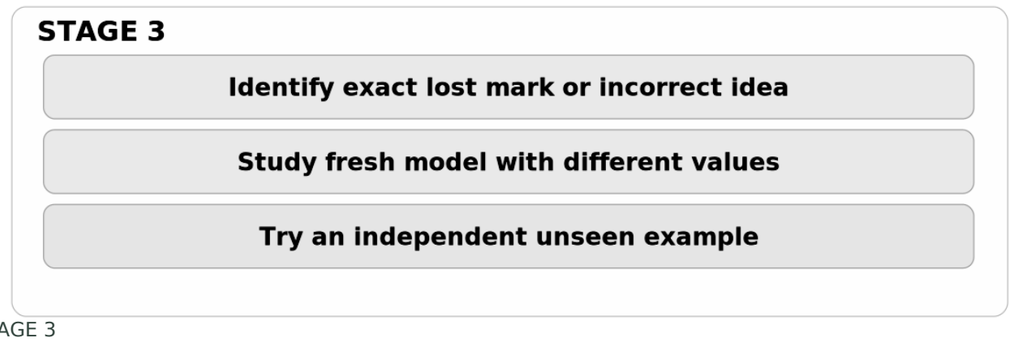
STAGE 1

New Learning



STAGE 2

New Learning



STAGE 3

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Show all steps for 8-bit two's complement −6.

Use exactly eight bits; inversion alone represents the wrong number.



WEDO · MODEL

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

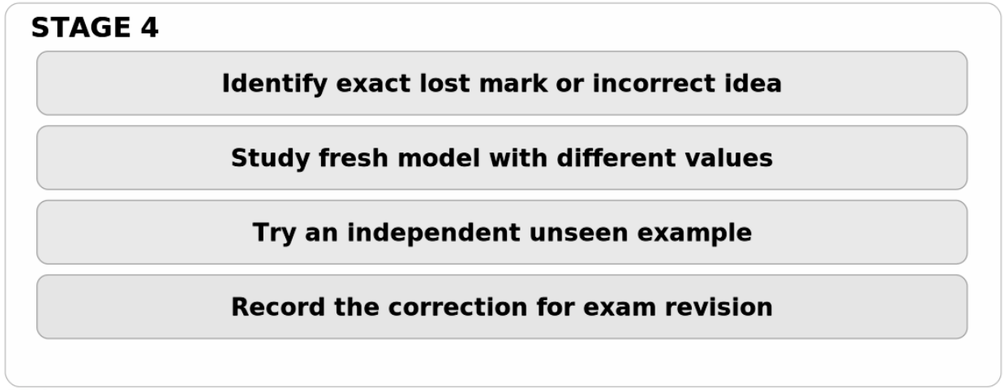
Guided 1 — CHECK

Explain the method
Use exactly eight bits; inversion alone represents the wrong number.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



STAGE 4

I DO — WORKED EXAMPLE

Follow the worked method

A sequential fetch starts PC=120. Explain what MAR, MDR and CIR do.

Explain the method

MAR holds address, MDR transfers contents, and CIR holds instruction for decoding.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Copying only the correct result proves a misconception has been resolved.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Student(ID,Name) contains (1,Ada),(2,Ben). Mark(ID,Score) contains (2,80). Write an INNER JOIN using Student.ID=Mark.ID to select Name and Score, and state the output row. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A sequential fetch starts PC=120. Explain what MAR, MDR and CIR do.

MAR holds address, MDR transfers contents, and CIR holds instruction for decoding.



IDO · MODEL

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Show all steps for 8-bit two's complement −6.
Source code / input:
00000110 → invert11111001 → add1 11111010.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

In a simple sequential fetch-decode-execute cycle, PC initially holds 70 and memory location 70 contains the instruction LOAD 25. State MAR, MDR, CIR and the next PC value, and explain one action of the control unit. Assume no branch or interrupt. [6]

Question 2

Convert −18 to eight-bit two's complement and explain sign. [6]